Unit 1 – Using Objects and Methods
Lesson 8: Math Class



What is the Math Class?

- The `Math` class is part of the `java.lang` package that automatically gets imported in every Java program.
- The `Math` class contains `static` methods, which mean that instead of using an object to call the method, we use the class name before the dot notation.
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- There are MANY math methods you can use. We will focus on the five that are included on the AP exam.

The following are the five `Math` methods detailed in the Java Quick Reference:

Math Methods	*What it does*	*Example*
`int abs(int x)`	Returns the absolute value of an int value	`Math.abs(-7) =`
`double abs(double x)`	Returns the absolute value of a double value	`Math.abs(5.67) =`
`double pow(double b, double e)`	Returns the value of the first parameter raised to the power of the second parameter	`Math.pow(4.5, 2) =`
`double sqrt(double x)`	Returns the positive square root of a double value	`Math.sqrt(25) =`
`double random()`	Returns a double value greater than or equal to 0.0 and less than 1.0	`Math.random() =`

Examples: What is printed to the screen when the following code segments are executed?

Code Segment	Console

`int x = 9, y = 2;`
`double answer1 = Math.abs(y - x);`
`System.out.println(answer1);`	

`int x = 9, y = 2;`
`double answer2 = Math.pow(x, y);`
`System.out.println(answer2);`	

`int x = 9, y = 2;`
`double answer3 = Math.sqrt(x) * y;`
`System.out.println(answer3);`	

Random Numbers

- The `Math.random()` method produces a random decimal between 0 and 1
- Random numbers can be very useful in all sorts of applications, but often we want a random integer
- Using some smart math, we can transform this random decimal into a random integer

Example: What are the possible values that could be generated with the following code segments?

Code Segment	Possible number range
`Math.random() * 10`	
`Math.random() * 100`	
`(int)(Math.random() * 10)`	
`(int)(Math.random() * 100)`	

`(int)(Math.random()*10) + 1`	
`(int)(Math.random()*100) + 8`	

Using the logic we developed above, there is a generic code segment we can use to randomly generate an integer number between two values:

```
int randNumber = (int) (Math.random() * range) + startingValue;
```

- _____ represents the number of values between the two endpoints (think about if you want the endpoint included too!)
- _____ represents the first integer number you want the random number to potentially be

Example

1) Write the code to generate a random number between 15 and 35, inclusively.

2) Write the code that simulates rolling two dice and calculating the sum.